

Have a Cow, Man (2010 Version)

Dissertation

Presented in Partial Fulfillment of the Requirements for the Degree
Doctor of Philosophy in the Graduate School of The Ohio State
University

By

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Graduate Program in Department of Nuclear Engineering

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2026

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Abstract

Field-programmable gate arrays (FPGAs) are increasingly attractive for nuclear instrumentation and control (I&C) because they offer deterministic hardware execution, reconfigurability, and high computational efficiency in compact, low-power platforms. However, their use in reactor-relevant environments is limited by radiation-induced effects such as single-event upsets and related functional disturbances. Conventional qualification campaigns can also be expensive and inefficient because they depend on scarce beam time, custom setups, and substantial external instrumentation. This work proposes a low-overhead irradiation-test methodology for commercial off-the-shelf SRAM-based FPGAs that emphasizes on-chip benchmarking, high-rate telemetry, and application-relevant fault observability.

The methodology combines two complementary workloads executed on the FPGA under identical beam conditions. The first is a flip-flop scan-farm benchmark that stresses a large population of explicitly instantiated registers and produces time-tagged mismatch events suitable for upset-rate and effective cross-section extraction. The second is a closed-loop digital control benchmark based on an I-PD controller coupled to a digital two-node thermal plant. Together, these workloads provide both fabric-level sensitivity information and application-level performance metrics, enabling comparison between raw upset behavior and functional degradation during

irradiation. Time-tagged telemetry from both workloads is streamed over UART for post-processing and comparison across test conditions.

Planned irradiations will be performed at The Ohio State University Research Reactor Fast Beam Facility, which provides a collimated fast-neutron beam with selectable gamma and thermal-neutron suppression. The intended configuration uses a bismuth gamma filter, a lead shutter, and an external cadmium filter to establish a reactor-relevant fast-neutron environment while suppressing unwanted beam components. Under the selected configuration, the facility characterization indicates a representative fast-neutron flux of approximately $2.6 \times 10^5 \text{ cm}^{-2} \text{ s}^{-1}$ with an associated gamma dose rate of about 50 mR/h.

The digital control workload is implemented entirely on-chip using signed fixed-point arithmetic and includes a discretized two-node thermal plant, measurement lag, actuator dynamics, filtered derivative action, and anti-windup. This structure allows radiation-induced perturbations to be observed directly in control-relevant variables such as tracking error, control effort, saturation behavior, and output response. In parallel, the scan-farm benchmark repeatedly executes LOAD, HOLD, and READ phases across 55 scan chains of 512 flip-flops each, corresponding to 28,160 registers, with a PRBS31 stimulus source and embedded signature-based checking.

The goal of this work is to establish a repeatable and transferable test approach that reduces setup overhead, improves statistical confidence during limited beam allocations, and better reflects realistic FPGA workloads in nuclear environments. If successful, this combined fabric-stress and application-benchmark methodology could provide a scalable path for lower-cost comparative radiation testing of candidate FPGA platforms for reactor-relevant service.

This is dedicated to the one I love ... la la la ...

Acknowledgments

I thank everyone who has ever had a cow. . . .

In reality, this is the only page of the dissertation which the author has full control of. You can write anything you want here, and no one can tell you it's wrong (except if the margins don't line up!!!!).

Vita

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B. Simpson “Milking a Cow”. *Journal of Dairy Science*, 00(2):277–287, Feb. 1900.

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Chapter 1: Introduction

Nuclear power occupies a singular position among low-carbon energy sources: it is the only technology that delivers firm, dispatchable, high-density electricity without direct greenhouse-gas emissions at a scale already proven across decades of operation. Yet the environments that make nuclear power possible — intense neutron and gamma fields, high temperatures, and aggressive chemical coolants — are precisely the environments that destroy the instruments needed to monitor and control it. Reliable radiation monitoring is not incidental to safe reactor operation; it is the primary mechanism by which operators verify that every barrier to fission-product release remains intact. As the existing fleet ages and a generation of advanced and small modular reactors (SMRs) moves from blueprint to concrete, the instrumentation technologies that have served the industry for half a century are confronting hard limits in both technical performance and economic viability. This dissertation investigates two complementary approaches to resolving that tension: the development of perovskite semiconductors as radiation-responsive detector materials, and the implementation of radiation-tolerant digital monitoring systems built on commercially available field-programmable gate arrays (FPGAs). Taken together, these efforts span the full stack from detector material to deployable system and aim to demonstrate

that next-generation nuclear instrumentation can be both technically capable and cost-effective.

1.1 Nuclear Power and the Global Energy Imperative

The scale of the decarbonization challenge facing the global electricity system makes any realistic pathway to net-zero heavily dependent on nuclear energy. At the COP28 climate summit in December 2023, twenty-two nations signed a declaration to triple global nuclear generating capacity from approximately 375 GW(e) to more than 1,125 GW(e) by 2050; that coalition has since grown to thirty-three countries [?]. The International Energy Agency’s net-zero pathway requires nuclear capacity to reach approximately 1,079 GW(e) by mid-century, demanding that annual investment in the sector roughly double from its current level of \$65–75 billion to \$120–150 billion per year by 2030 [?]. The International Atomic Energy Agency (IAEA) has revised its projections upward for five consecutive years, with its 2025 high-case scenario projecting 992 GW(e) by 2050 [?].

The existing global fleet comprises approximately 408 operating reactors in thirty-one countries, with a combined net capacity that reached a record 369 GW(e) in 2024 [?]. Nuclear electricity generation also set an all-time high of 2,667 TWh in 2024, narrowly surpassing the 2006 record, though nuclear’s share of global electricity has fallen to approximately 9% as variable renewables have expanded at a far faster pace [?]. The fleet’s average age of 32.4 years — with 141 units exceeding 41 years of operation — makes life extension, instrumentation modernization, and new construction simultaneously urgent priorities.

The advanced reactor pipeline is the most active in decades. In the United States, TerraPower’s 345 MW(e) Sodium sodium-cooled fast reactor received its NRC construction permit in March 2026, the first commercial non-light-water reactor approval in over forty years [?]. Kairos Power’s Hermes fluoride-salt-cooled test reactor began nuclear safety-related construction in 2025 [?], and X-energy’s Xe-100 pebble-bed high-temperature gas-cooled reactor is under development for an industrial host site with over \$1.2 billion in DOE cost-share [?]. Among light-water SMRs, GE-Hitachi’s BWRX-300 is under construction at Ontario’s Darlington site and has received a \$400 million DOE award for deployment at TVA’s Clinch River site [?]. Each of these programs introduces novel coolant chemistries, fuel geometries, and multi-module control architectures that existing monitoring technologies were never designed to address.

1.2 The Regulatory Foundation for Radiation Monitoring

Continuous radiation monitoring across neutron flux, gamma dose, primary coolant activity, and effluent pathways is mandated by the regulatory foundations of nuclear operation, not merely encouraged by best practice. The NRC’s General Design Criteria, codified in 10 CFR 50 Appendix A, establish this requirement at the highest level of the licensing framework. GDC 13 requires instrumentation capable of monitoring all variables that affect the reactor core, coolant pressure boundary, and containment integrity; GDC 64 mandates monitoring of the containment atmosphere, recirculation spaces, and effluent pathways for radioactivity; and GDC 19 requires that control room operators maintain safe-shutdown capability with adequate instrumentation even under accident conditions [?]. The IAEA mirrors these requirements through SSR-2/1 and provides detailed I&C system guidance in SSG-39 [?, ?].

The technical scope of these monitoring requirements is extraordinary. Neutron flux measurement alone spans approximately twelve decades — from a few counts per second during refueling to approximately $10^{15} \text{ n cm}^{-2} \text{ s}^{-1}$ at full power — requiring three overlapping detector subsystems. Source-range monitors based on proportional counters or fission chambers operate in pulse-counting mode across the lowest six to seven decades during startup and shutdown. Intermediate-range monitors employ compensated ionization chambers in Campbell mode across approximately eight to ten decades. Power-range monitors using uncompensated boron-lined ionization chambers cover the upper two decades and are calibrated against secondary-side heat balance. Each of these subsystems must meet independent safety-channel redundancy requirements, typically two to four independent channels per range [?].

Beyond neutron flux, plants deploy area gamma monitors throughout containment, auxiliary buildings, fuel handling areas, and spent fuel pools to enforce occupational dose limits under 10 CFR 20. On-line gamma spectroscopy of the primary coolant monitors fission-product signatures that indicate fuel-cladding breaches. Containment atmosphere monitors classified as Category 1 under Regulatory Guide 1.97, Revision 5, must survive combined temperature, pressure, radiation, and chemical-spray environments for the full duration of a design-basis accident [?]. Stack and liquid effluent monitors verify compliance with 10 CFR 50 Appendix I dose objectives, while the Radiological Environmental Monitoring Program tracks off-site air, water, and vegetation samples under RG 4.1.

The Fukushima Daiichi accident of March 2011 provided a severe demonstration of the consequences when monitoring fails. The tsunami rendered 23 of 24 on-site radiation monitoring stations nonfunctional. The subsequent loss of all AC power and

battery depletion eliminated reactor vessel pressure, temperature, and water-level indication, forcing operators to manage three simultaneous core-melt progressions with essentially no real-time process data [?, ?]. Spent fuel pool instrumentation was unavailable for nearly two weeks. The NRC’s post-Fukushima response, including Order EA-12-051 requiring backup-powered spent fuel pool level instrumentation, the FLEX diverse-coping strategy described in NEI 12-06, and the addition of Type F severe-accident management variables to RG 1.97 Rev. 5, underscored that reliable monitoring is not a compliance formality but the operational foundation of accident management.

1.3 Limitations of Current Radiation Detector Technologies

The detector materials deployed across the nuclear industry each carry critical limitations that constrain their use in existing plants and threaten to preclude their adoption in advanced reactor environments. High-purity germanium (HPGe) remains the gold standard for gamma spectroscopy, achieving better than 0.2% full-width at half-maximum energy resolution at 662 keV. However, its narrow bandgap of 0.67 eV mandates cryogenic operation below 120 K. Liquid-nitrogen dewars are impractical for unattended or remote monitoring; mechanical coolers add substantial cost, complexity, and microphonic noise that degrades resolution at low energies. HPGe is therefore confined almost exclusively to laboratory and assay applications rather than continuous in-plant monitoring [?].

Silicon detectors dominate charged-particle detection and fast-neutron spectrometry in research settings but suffer irreversible displacement damage above approximately 10^{14} – 10^{15} n_{eq} cm^{−2} fluence, with charge-collection efficiency falling toward 50%

— a threshold easily surpassed in the lifetime of a power-reactor instrument [?]. Silicon’s low atomic number ($Z = 14$) also provides poor photoelectric stopping power for gamma-ray energies above a few hundred keV, making it poorly suited to the mixed neutron-gamma fields present throughout a reactor plant.

Scintillators constitute the most widely deployed class of gamma-detection instruments and are valued for their ability to produce measurable light from a wide range of ionizing radiations. Sodium iodide thallium-doped [NaI(Tl)] crystals, the workhorse material, achieve approximately 6–7% energy resolution at 662 keV but are highly hygroscopic, mechanically fragile, and must be coupled to photomultiplier tubes that are sensitive to magnetic fields, fragile under shock and vibration, and require high-voltage supplies of 800–1200 V [?]. Advanced scintillators such as LaBr₃(Ce) achieve approximately 3% resolution but contain intrinsic radioactivity from ¹³⁸La that creates a flat background continuum complicating low-level measurements. Organic scintillators, including plastic scintillators and stilbene crystals, are well established for fast neutron detection through proton-recoil interactions, but their low effective atomic number renders them essentially transparent to gamma rays, requiring either active discrimination electronics or companion gamma-shielding that adds mass, cost, and complexity.

Cadmium zinc telluride (CZT) represents the most mature room-temperature semiconductor alternative, offering energy resolutions approaching 0.7% at 662 keV in pixelated geometries with full single-carrier-transport correction. However, CZT crystal growth requires extremely precise control of stoichiometry and vapor-phase conditions during directional solidification: high cadmium vapor pressure promotes vacancy defects, tellurium inclusions limit boule diameter, and critical shear stresses

of only 0.3 MPa make the material mechanically fragile during processing [?]. Production costs remain high, supply is constrained, and the situation deteriorated significantly in 2025 when the People’s Republic of China imposed export controls on germanium, gallium, and related critical materials including cadmium telluride compounds.

For thermal neutron detection, the ^3He tube has been the standard technology since the 1950s by virtue of its high neutron absorption cross-section (5330 barns at 0.025 eV), excellent gamma discrimination, and compatibility with conventional pulse-counting electronics. The supply of ^3He — produced exclusively as a decay product of tritium from the nuclear-weapons stockpile — was recognized as critically constrained following the post-September 11 deployment of approximately 1,300 radiation portal monitors for border-security applications. Annual demand reached roughly 70,000 L against a production rate of only approximately 8,000 L yr⁻¹, driving prices from approximately \$100 L⁻¹ to over \$2,000 L⁻¹ and prompting DOE to begin rationing allocation in 2009 [?]. The episode exposed the fragility of depending on a single non-renewable detection medium and has accelerated development of alternative neutron detectors using ^{10}B -lined tubes and $^6\text{LiF}:\text{ZnS}$ scintillator screens.

Halide perovskites have emerged as a compelling alternative class of radiation-responsive materials. Their high effective atomic number — CsPbBr_3 has $Z_{\text{eff}} = 65.9$, compared to 50.2 for CZT — provides excellent photoelectric stopping power for gamma radiation [?]. Single-crystal MAPbBr_3 and CsPbBr_3 devices have demonstrated mobility-lifetime products of up to $8 \times 10^{-3} \text{ cm}^2 \text{ V}^{-1}$ and energy resolutions approaching 1.4% at 662 keV [?], approaching CZT performance without cryogenic cooling or Bridgman growth. A distinctive self-healing mechanism — in which the

soft perovskite lattice allows radiation-displaced ions to migrate back toward equilibrium positions at room temperature — has been confirmed under proton irradiation to $10^{13} \text{ p cm}^{-2}$ and extended gamma exposures [?]. The methylhydrazinium lead chloride (MHyPbCl₃) composition investigated in this work introduces a high-hydrogen-density 3D perovskite framework with a hydrogen density of $3.79 \times 10^{22} \text{ cm}^{-3}$ comparable to organic scintillators, enabling direct fast-neutron conversion through proton recoil without requiring a separate moderator or conversion layer [?].

Significant challenges remain. Ion migration under applied electric fields and elevated humidity drives hysteresis in current-voltage characteristics and accelerates device failure at the biases required for thick-crystal direct-conversion operation. Long-term stability data at power-reactor fluence levels (10^{14} – $10^{15} \text{ n cm}^{-2}$) remain absent from the literature. Lead toxicity presents regulatory and environmental considerations for deployed instruments. These challenges motivate the stabilization strategies — fluorinated surface passivation, hexagonal boron nitride composites, and vacuum operation — investigated in the present work.

1.4 The Economics of Nuclear Instrumentation

Cost overruns in nuclear construction have become the defining obstacle to expansion in liberalized electricity markets. The two AP1000 units completed at Plant Vogtle in Georgia cost approximately \$35 billion against an original estimate of \$14 billion [?]. France’s Flamanville 3 EPR reached €23.7 billion — seven times its original budget — after seventeen years of construction. The UK’s Hinkley Point C has escalated to an estimated £35 billion in 2015 prices against an initial projection of approximately £18 billion [?]. New nuclear leveled cost of electricity in the United

States ranges from approximately $\$110 \text{ MWh}^{-1}$ (EIA) to $\$180 \text{ MWh}^{-1}$ (Lazard), compared to approximately $\$55 \text{ MWh}^{-1}$ for utility-scale solar and $\$40 \text{ MWh}^{-1}$ for onshore wind. The existing U.S. fleet operates far more competitively — NEI reports a 2023 average total generating cost of $\$31.76 \text{ MWh}^{-1}$, the lowest on record [?] — but twelve commercial reactors have closed since 2013 due to competitive pressure, and emergency support programs including a $\$6$ billion Civil Nuclear Credit allocation have been required to prevent further premature retirements [?].

Instrumentation and control systems, while representing an estimated 5–10% of total plant capital cost, exert a disproportionate influence on both construction timelines and long-term operating expenses. A typical large light-water reactor contains approximately 10,000 sensors, 5,000 km of I&C cabling, and hundreds of electronic cabinets. Nuclear-grade quality assurance requirements under 10 CFR 50 Appendix B impose a cost premium that pervades the entire supply chain: components manufactured to identical functional specifications have been documented at price ratios of 4:1 or greater between nuclear-qualified and commercial-grade procurement, driven by QA documentation requirements, environmental qualification testing, and the severe scarcity of nuclear-qualified suppliers [?]. Dominion Energy’s comprehensive digital I&C modernization across four units at Surry and North Anna is budgeted at $\$5.45$ billion as part of subsequent license renewal [?]. The DOE Light Water Reactor Sustainability (LWRS) Program has explicitly cautioned that “wide-scale modernization of legacy I&C systems could prove to be cost-prohibitive unless implemented to enable significant business innovation.”

The cost of radiation-hardened digital electronics amplifies this problem at the component level. Purpose-built radiation-hardened FPGAs — the Microchip RTG4,

AMD/Xilinx XQRKU060, and their equivalents — carry price tags of tens of thousands to hundreds of thousands of dollars per device, a premium of $10\times$ to $100\times$ over commercial equivalents that cost \$10–500 [?]. The radiation-hardened FPGA market, valued at \$465.5 million in 2024, serves primarily space and defense applications where unit volumes are small and alternatives do not exist. These devices trail commercial FPGAs by two to three process generations: the Microchip RTG4 uses a 65 nm flash-based process while commercial devices operate at 3–5 nm, a gap representing multiple orders of magnitude in logic density, performance, and power efficiency [?, ?]. Long lead times of 6–18 months, ITAR export restrictions, and the absence of advanced intellectual property cores compound the challenge. For SMRs targeting a levelized cost of electricity of $\$60\text{--}90\text{ MWh}^{-1}$, this cost structure is untenable: I&C requirements do not scale down proportionally with reactor output, making instrumentation a higher fraction of total cost in smaller plants. The DOE’s GEMINA program has set an aspirational target of reducing advanced reactor operations and maintenance cost from $\$23\text{ MWh}^{-1}$ to $\$2\text{ MWh}^{-1}$ through autonomous operations and advanced diagnostics — a goal that cannot be achieved without fundamentally cheaper and more capable sensing and processing hardware [?].

The alternative to purpose-built rad-hard electronics is software-defined radiation tolerance using commercial off-the-shelf FPGAs with designer-inserted mitigation. Triple modular redundancy (TMR) triplicates critical logic with majority voting to correct single-event upsets in real time; configuration memory scrubbing periodically corrects bit-flip errors in SRAM-based FPGAs before they propagate; and partial dynamic reconfiguration enables real-time repair of corrupted fabric regions. CERN’s two-year study of Kintex-7 FPGAs under mixed-field radiation found no permanent

operational failures to 300 krad total ionizing dose with TMR and scrubbing [?]. Montana State University’s RadSat program demonstrated COTS FPGA-based radiation tolerance at “a cost 100× lower than existing radiation-hardened computers.” A TMR-with-spares architecture for nuclear reactor protection system logic has been validated under cyclotron irradiation [?]. These results suggest that COTS FPGAs with mature mitigation strategies can meet nuclear monitoring requirements at a fraction of the cost of purpose-built rad-hard devices, provided the mitigation architecture is co-designed with the application from the outset.

1.5 The Gap: An Integrated Materials-to-Systems Approach

Nuclear instrumentation research has historically operated in disciplinary silos. Detector material development, device fabrication, signal processing electronics, and system-level integration have each been pursued as largely independent research tracks. The DOE’s Nuclear Energy Enabling Technologies (NEET) Advanced Sensors and Instrumentation (ASI) program, which has funded more than 43 projects since FY 2011 with a combined investment exceeding \$36 million, organizes its solicitations into discrete categories covering sensors, control systems, communications, and data analytics [?]. Conference proceedings from the ANIMMA international instrumentation symposium similarly separate detector development sessions from electronics sessions, with system integration papers being comparatively rare.

The cost of this fragmentation is measurable. A 2024 Nature Communications study on CsPbBr₃ alpha-particle detectors achieved 6.9% energy resolution with commercial laboratory electronics and 1.1% energy resolution with a custom-designed readout ASIC matched to the perovskite’s specific carrier-transport characteristics

— a sixfold improvement achieved entirely through electronics optimization rather than material improvement [?]. Yet the authors note that “perovskite semiconductors combined with corresponding electronic designs have not been largely explored so far.” Meanwhile, in-reactor perovskite dosimetry demonstrations have relied on bench-top source-measurement units costing thousands of dollars, because no deployable, radiation-tolerant readout system exists for these materials. The fundamental economic argument for perovskites — low-cost solution processing, room-temperature operation, and material costs orders of magnitude below CZT — is negated if the readout system costs remain at the level of conventional rad-hard electronics.

This dissertation addresses that gap through three integrated areas of investigation. First, at the materials level, perovskite compositions spanning 3D and 2D structures and ranging from high-energy photon monitoring to fast neutron direct conversion are synthesized, characterized, and evaluated for radiation response. Second, at the device level, these materials are developed into functional detectors, with each device architecture matched to a specific radiation detection application. Third, at the system level, a radiation-tolerant FPGA-based monitoring platform is developed and validated through digital plant simulation, providing the processing infrastructure necessary to deploy perovskite-based detectors in a nuclear environment without reliance on purpose-built rad-hard components. The connecting thesis is straightforward: novel radiation-responsive materials and cost-effective radiation-tolerant systems must be developed together if the resulting instrumentation is to be technically capable and economically viable for advanced reactor deployment.

1.6 Research Objectives and Dissertation Organization

The research presented in this dissertation is organized around three complementary objectives aligned with the materials-to-systems hierarchy described above.

The first objective is to characterize the radiation response of halide perovskite materials across a range of compositions, device architectures, and radiation types relevant to nuclear monitoring. This encompasses gamma-photon response from Cs-137 and Co-60 sources, X-ray sensitivity as a function of applied bias, alpha-particle spectroscopy, and the material instability mechanisms — ion migration, humidity-dependent breakdown, and charge trapping — that currently limit perovskite detector lifetimes.

The second objective is to develop and demonstrate perovskite-based radiation detection devices optimized for specific applications in the nuclear fuel cycle: high-energy photon monitoring for post-detonation forensics, direct-conversion X-ray detection, alpha-particle spectroscopy for thermal neutron sensing feasibility, and direct fast-neutron conversion using the hydrogen-rich MHyPbCl_3 composition. Each device is characterized through electrical transport measurements, Monte Carlo simulation, and experimental irradiation at a research reactor facility.

The third objective is to implement and validate a radiation-tolerant FPGA-based digital monitoring system suitable for nuclear plant instrumentation applications. This includes design and testing of a fault-tolerant flip-flop scrubbing register (FFSR) architecture, integration of TMR voter logic, and validation of the combined system against a digital plant simulation environment representing nuclear I&C functions.

The remainder of this dissertation is organized as follows. Chapter 2 provides the background necessary for each area of investigation, encompassing perovskite crystal structure and radiation interaction physics, ion migration mechanisms, radiation effects on digital electronics, and the taxonomy of fault-tolerant design approaches. Chapter 3 presents the characterization of $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}$ perovskite solar cells under extended gamma irradiation. Chapter 4 addresses the voltage-induced instability problem in two-dimensional perovskite detectors and the 5F-PEAI surface passivation strategy developed to overcome it. Chapter 5 describes the MAPbI_3 -hBN composite detector and its characterization for X-ray and alpha-particle detection. Chapter 6 presents the MHyPbCl_3 fast neutron detector, from crystal synthesis through beam-facility testing and gamma discrimination analysis. Chapter 7 describes the radiation-tolerant FPGA monitoring system, including the FFSR design, TMR integration, and digital plant simulation validation. Chapter 8 provides overall conclusions and recommendations for future work.

Chapter 2: Background and Prior Work

Chapter 3: Testing of an Organic Metal Halide Perovskite for Fast Neutron Detection

Chapter 4: Contributions and Future Work

We have successfully shown in Chapter ?? how the problems of having a cow can be solved. A lot of work can still be done in this field so that better and bigger cows can be had. This is a matter which can be pursued by those weeny grad students who have just started their Ph.D.

In reality, this document has been prepared by Manas Mandal and Al Fencel to assist people in using the `osudissert96` class for $\text{\LaTeX}$ to generate masters theses and doctoral dissertations at the Ohio State University. The relevant files are available from the Department of Computer and Information Science or Electrical Engineering. Also, ask around your department; the files may have already migrated to a computer in your home department.

Keep in mind that this document *is not intended to teach you $\text{\LaTeX}$* nor should it be used as such. Lamport's book [?] is a very good introduction to the wonderful world of $\text{\LaTeX}$ and it would be well worth your while to purchase that book!

It is our sincere hope that the use of this document and the related style files will help you in producing your thesis and/or dissertation. We wish you "Good Luck"!

Appendix A: The Data on Cows

This is the data that was used to produce the table in Table ?? . In 1990, 57 people had cows. In 1991, 80 people had cows. In 1992, 199 people had cows.

Appendix B: Important commands defined in osudissert96

The following is a list of all commands available in osudissert96.

```
\author{First Middle Last}
\title{The title of the thesis}
\authordegrees{degree1, degree2}
\unit{Department of Whatever The Name Is}
\degree{Doctor of Philosophy}
\committee{Dissertation}
\advisorname{Name of advisor} % Possible usage "Prof. Big Dude"
\member{Name of committee member}

\thesis % this makes it a thesis rather than a dissertation,
        % similar to the documentclass option [masters] or [ms].

\maketitle

\disscopyright % or \blankpage

\begin{abstract}
\end{abstract}

\begin{externalabstract}
\end{externalabstract}

\dedication{This is dedicated to \ldots}

\begin{acknowledgements}
\end{acknowledgements}

\begin{vita}
```

```

\dateitem{Important Date}{ Why its important}

\begin{publist}
\researchpubs
\pubitem{Bibliography item (from BibTeX?)}

\instructpubs
\pubitem{Bibliography item (from BibTeX?)}
\end{publist}

\begin{fieldsstudy}
\majorfield*           % which uses \unit above
% \majorfield{Your Major Field}
\begin{studieslist}
\studyitem{Topic 1}{Prof.\ 1}
\studyitem{Topic 2}{Prof.\ 2}
\studyitem{Topic 3}{Prof.\ 3}
\end{studieslist}
%% Note:  If there were only one field of study, the following list
%%         would best be done using the following command:
%% \onestudy{Only Topic}{Only Professor}
\end{fieldsstudy}

\end{vita}

```

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